SOC ANALYST HOME LAB PROJECT

SOC Monitoring Dashboard, Incident Investigation & Final Validation

Splunk SIEM • Windows Security Monitoring • Alert Triage • Incident Analysis

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Project Role: SOC Analyst / Lab Owner

Personal Controlled Cybersecurity Home Lab | Completed October 2026

Project type	Personal SOC Analyst home lab
Environment	VMware Workstation — Bridged networking
SIEM	Splunk Enterprise 10.4.3 on Ubuntu Server
Monitored endpoint	Windows 10 — DESKTOP-4AMTH10
Analyst workstation	Kali Linux / Firefox
Log transport	Splunk Universal Forwarder → TCP 9997 → Splunk Enterprise
Project completion checkpoint	Dashboard, alert validation, incident investigation, disposition, and final health validation
Final validation date	October 3, 2026
Project owner	Norbert Endoh — SOC Analyst / Lab Owner

Portfolio accuracy statement

This document describes a controlled home-lab implementation, not a production enterprise SOC. All security activity described as testing was intentionally generated in the user's own lab environment. The workflow mirrors core SOC practices at small scale.

1. Executive Summary

This final project checkpoint documents the monitoring, alerting, investigation, and closure stages of the Splunk SOC home lab. The lab progressed from a working Windows-to-Splunk telemetry pipeline into an operational analyst workflow: a Windows security dashboard was built, scheduled detections were enabled, controlled failed-logon activity triggered an alert, the alert was investigated through authentication and process telemetry, and the incident was dispositioned as a true-positive detection generated by an authorized lab test.

The final health check confirmed that Windows Security telemetry was still arriving in Splunk, the detection alerts remained enabled, and the SOC dashboard continued to display current authentication and process activity.

2. Relationship to Earlier Project Documents

This is Document 3 in the project series. It intentionally complements, rather than repeats, the earlier documentation.

- Document 1 — Foundation, Build & Early SOC Analysis: architecture, bridged networking, Ubuntu Splunk server, Universal Forwarder, ingestion pipeline, and early Windows event analysis.
- Document 2 — Detection Engineering Use Case 01: repeated failed Windows logons, Event ID 4625, SPL development, field normalization, scheduled alerting, validation, and suppression/tuning.
- Document 3 — This document: SOC dashboard, alert validation, triggered-alert investigation, authentication/process correlation, incident disposition, and final project health validation.

3. SOC Windows Security Monitoring Dashboard

A dedicated Splunk dashboard was created to provide a single analyst view for Windows authentication, account-management, privileged-logon, process-creation, and PowerShell-related telemetry. The dashboard uses a time-range input and investigation tables so the analyst can move from high-level counts to event-level evidence.

- Failed Windows Logons — Event ID 4625
- Successful Windows Logons — Event ID 4624
- Windows Authentication Activity Over Time — Event IDs 4624 and 4625
- Failed Windows Logons by User
- Top Windows Security Event IDs
- Failed and Successful Logon Investigation Details
- Windows Account Management Events — Event IDs 4720–4726
- Privileged/Special Logon Activity — Event ID 4672
- Windows Process Creation Activity — Event ID 4688
- PowerShell Execution Activity — Event ID 4688
- Process Creation Investigation Details
- Account creation/deletion and local-group membership change panels

Final dashboard validation used the Last 24 hours time range. At validation time, the dashboard displayed 6 failed Windows logons and 165 successful Windows logons. The failed-logon investigation table preserved the six controlled labuser failures, while the successful-logon table showed the subsequent interactive logons.

4. Detection Alert Validation

The final alert review confirmed five enabled Splunk alerts:

- Administrator Group Membership Addition
- Repeated Failed Windows Logons - 3 in 5 Minutes
- Windows Administrator Group Membership Change
- Windows Local Account Creation
- Windows PowerShell Execution Detection

The primary incident use case remained the repeated-failed-logon alert. It was configured to identify three or more failed Windows logons for the same target user within a five-minute bucket, run every five minutes over the last fifteen minutes of data, add results to Triggered Alerts, and suppress duplicate triggering for the same target_user for fifteen minutes.

5. Triggered Alert Evidence

Alert	Repeated Failed Windows Logons - 3 in 5 Minutes
Severity	High
Trigger history	Scheduled alert fired successfully
Trigger time observed	2026-10-03 18:10:04 UTC
Target user	labuser
Failed logons	3
Source address	127.0.0.1
Workstation	DESKTOP-4AMTH10
Detection bucket	5 minutes

The triggered result proved that the complete detection chain worked automatically: controlled Windows authentication failures were logged, forwarded, searched by the scheduled detection, and surfaced as a triggered alert.

6. SOC Incident Investigation

6.1 Authentication Timeline

The analyst correlated failed Event ID 4625 records with successful Event ID 4624 records for labuser. The most relevant sequence was:

Time	Event ID	Meaning	Source	Logon Type
18:05:06	4625	Failed logon	127.0.0.1	2
18:05:12	4625	Failed logon	127.0.0.1	2
18:05:17	4625	Failed logon	127.0.0.1	2
18:05:32	4624	Successful logon	127.0.0.1	2

Logon Type 2 represents an interactive/local logon. The sequence therefore showed three failed attempts followed shortly by a successful interactive authentication on the monitored endpoint.

6.2 Process Activity Review

Event ID 4688 process-creation telemetry was reviewed to determine what occurred in the labuser session after authentication. The initial process search was intentionally broad and returned substantial Windows/service noise. The investigation was then narrowed to labuser.

- Observed user-session activity included taskhostw.exe, Microsoft Edge processes, Microsoft 365 Copilot, RuntimeBroker.exe, backgroundTaskHost.exe, and related Windows components.
- A targeted PowerShell search for labuser returned no results in the investigated 30-minute window.
- Splunk Universal Forwarder helper processes visible elsewhere in the dashboard were treated as monitoring-agent activity, not evidence that labuser launched PowerShell.

6.3 Investigation SPL

Authentication correlation:

```
index=* host="DESKTOP-4AMTH10" sourcetype="WinEventLog:Security" (EventCode=4624 OR
EventCode=4625)
| eval user=coalesce(TargetUserName, Account_Name)
| where user="labuser"
| table _time EventCode user Source_Network_Address Workstation_Name Logon_Type
| sort _time
```

Process isolation:

```
index=* host="DESKTOP-4AMTH10" sourcetype="WinEventLog:Security" EventCode=4688
| eval user=coalesce(SubjectUserName, Account_Name)
| where user="labuser"
| table _time user New_Process_Name Process_Command_Line
| sort _time
```

Targeted PowerShell check:

```
index=* host="DESKTOP-4AMTH10" sourcetype="WinEventLog:Security" EventCode=4688
| eval user=coalesce(SubjectUserName, Account_Name)
| where user="labuser"
| search New_Process_Name="*powershell.exe"
| table _time user New_Process_Name Process_Command_Line
| sort _time
```

7. Final SOC Incident Record

Incident title	Repeated Failed Windows Logons Followed by Successful Authentication
Affected account	labuser
Affected host	DESKTOP-4AMTH10
Source address	127.0.0.1
Relevant Windows events	4625, 4624, 4688
Severity	High
Status	Closed
Disposition	True Positive — Authorized/Benign Security Test
Containment	Not required
Escalation	Not required

Incident summary: Splunk detected three failed authentication attempts against labuser within the defined threshold. Investigation confirmed a subsequent successful interactive logon and reviewed associated process-creation telemetry. No labuser PowerShell execution was identified in the investigated 30-minute window. Because the activity was intentionally generated as part of the controlled home lab, the alert was a true-positive detection of the defined behavior but the incident itself was authorized and benign.

Production-response note: Similar behavior in a real environment would require validation of the account owner, source system/address, authentication context, post-login activity, and other surrounding telemetry before closure or escalation.

8. Final Technical Health Validation

Before closing the project, the monitoring pipeline, alerts, and dashboard were revalidated.

Telemetry check	PASS — 114 Windows Security events received in the final 15-minute validation window
Latest event observed	2026-10-03 18:34:02.410
Detection alerts	PASS — 5 alerts enabled
SOC dashboard	PASS — current Windows authentication/process telemetry displayed
End-to-end status	PASS — ingestion → detection → alerting → investigation → visualization

Final telemetry validation SPL:

```
index=* host="DESKTOP-4AMTH10" sourcetype="WinEventLog:Security"
| stats count as Total_Security_Events latest(_time) as Latest_Event
| convert ctime(Latest_Event)
```

9. Skills Demonstrated

- Splunk Enterprise SIEM deployment and use
- Splunk Universal Forwarder and Windows Security log ingestion
- SPL searching, filtering, statistics, and field normalization
- Windows Security Event IDs 4624, 4625, 4672, 4688, 4720–4726, 4732, and 4733
- Detection engineering and threshold-based alert logic
- Scheduled alerts, cron scheduling, severity, throttling, and suppression
- SOC dashboard development and investigation tables
- Alert triage and triggered-alert validation
- Authentication timeline correlation
- Process-creation investigation
- Incident disposition and closure documentation
- Final SIEM health validation and evidence-based reporting

10. Portfolio / Resume Project Description

SOC Analyst Home Lab — Splunk SIEM for Windows Security Monitoring, Detection & Investigation

- Built a segmented SOC home lab in VMware Workstation with Splunk Enterprise on Ubuntu Server, a Windows 10 monitored endpoint, Splunk Universal Forwarder, and Kali Linux as the analyst workstation.
- Collected and analyzed Windows Security telemetry in Splunk and created a multi-panel SOC dashboard covering authentication, privileged logons, account-management events, process creation, and PowerShell activity.
- Engineered and validated a repeated-failed-logon detection using Event ID 4625, scheduled alerting, per-user suppression, and controlled event generation.

- Investigated a triggered alert by correlating failed and successful authentication events with Event ID 4688 process activity, then documented the incident disposition and closure.
- Performed final health validation confirming active telemetry ingestion, enabled detections, and functioning dashboard visibility.

11. Interview Explanation

“I built a Splunk SOC home lab with a Windows 10 endpoint forwarding Security logs to Splunk Enterprise on an Ubuntu Server. I created a dashboard for Windows authentication and process monitoring and engineered a detection for three failed logons within five minutes. I validated the scheduled alert with controlled failed logins, investigated the triggered alert, correlated the failures with a subsequent successful interactive logon, and reviewed Event ID 4688 process activity for the user. I also checked specifically for PowerShell execution. Because the activity was intentionally generated in my lab, I closed it as a true-positive detection with an authorized/benign test disposition. Finally, I revalidated the ingestion pipeline, enabled alerts, and dashboard to prove the SIEM was still operating end to end.”

12. Project Completion Statement

The core SOC Analyst home-lab project is complete for this scope. The environment demonstrates a functioning Windows-to-Splunk telemetry pipeline, security monitoring dashboard, multiple enabled detection alerts, a validated scheduled detection, alert triage, event correlation, process investigation, incident disposition, and final operational validation.

PROJECT STATUS: COMPLETE